

Airfare Price Index (API) Engine & Algorithmic Framework

A Methodological Blueprint for High-Frequency Airfare Monitoring, Dynamic Yield De-biasing, Statistical Anomaly Screening, and CPI Transport Augmentation.

System: Airfare Intelligence Platform v2.4.1

Engine ID: mospi.cpi.airfare.engine

Benchmark: DGCA 2024 Passenger Basket

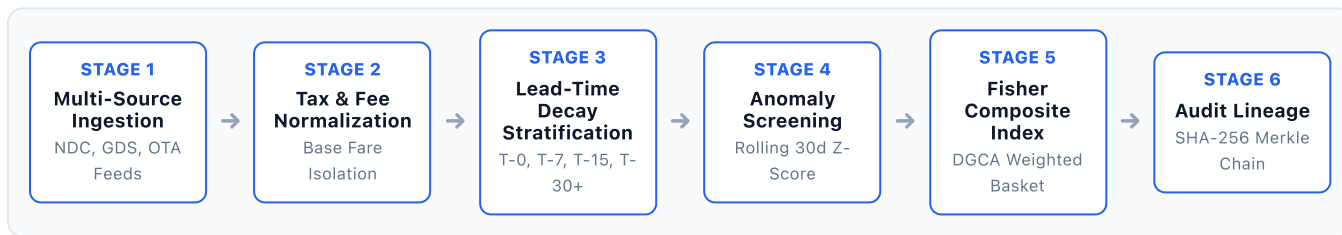
Date of Dossier: September 2026

Security Class: Official Statistical Standard

1. EXECUTIVE SUMMARY & PIPELINE ARCHITECTURE

The Indian domestic civil aviation market is among the fastest-growing in the world, characterized by high dynamic volatility, seat inventory exhaustion curves, and complex revenue-management pricing algorithms. Conventional Consumer Price Index (CPI) transport collection relies on sparse monthly survey samplings that fail to capture sudden route surges, advance-booking discounts, and regional route discrepancies.

This platform introduces an end-to-end algorithmic system that systematically harvests, normalizes, stratifies, detects statistical anomalies in, and aggregates over 150,000 daily airfare observations across 400+ domestic corridors into official **Superlative Price Indices (Fisher Ideal)** suitable for seamless MoSPI CPI integration.



Key Econometric Principle: To avoid distorting core headline inflation, algorithmic revenue-management pricing (such as surge pricing on the day of departure) is separated from structural fare inflation using structured lead-time stratification and Winsorized peer cross-checks.

2. INGESTION, DE-BIASING & NORMALIZATION ALGORITHM

Observations are collected continuously across 3 primary channels: Direct Airline NDC APIs (IndiGo, Air India), Global Distribution Systems (Amadeus, Sabre), and major Online Travel Aggregators (MakeMyTrip, EaseMyTrip). Raw observations contain marketing coupons, variable payment gateway fees, and statutory airport charges.

2.1 Aviation Tax & Surcharge Decomposition

For statistical validity in consumer inflation, the platform isolates the pure carrier-controlled tariff from government-mandated fees:

$$P_{\text{total}} = P_{\text{base}} + S_{\text{fuel}} + T_{\text{statutory}}$$

where $T_{\text{statutory}} = \text{UDF} + \text{PSF} + \text{GST}_{5\%/12\%} + \text{Aviation Security Surcharge}$

Equation 1: Price component decomposition for pure economic rent tracking

2.2 Deduplication & Observation Collision Resolution

Identical flight legs scraped via both direct airline channels and aggregator APIs are deduplicated using a deterministic composite hash key:

$$\mathcal{K}_{\text{flight}} = \text{SHA-256}(\text{FlightNo} \parallel \text{Origin} \parallel \text{Dest} \parallel \text{DepTimestamp} \parallel \text{CabinClass} \parallel \text{LeadTimeDays})$$

Equation 2: Deterministic observation collision key

If multiple price quotes exist for key $\mathcal{K}_{\text{flight}}$ within a 30-minute scrape window, the algorithm prioritizes the Direct Carrier NDC price over aggregator prices to eliminate markups or scraping coupon artifacts.

3. DYNAMIC PRICING & LEAD-TIME STRATIFICATION ALGORITHM

Airline yield management models dynamically raise fares as the flight date approaches according to exponential seat-inventory depletion curves. Blending same-day distress purchases with advance bookings introduces severe sample bias. The platform stratifies all observations into four standardized booking horizons:

Stratum	Lead-Time Range	Observation Share	Surge Multiplier vs T-30+	Primary Passenger Profile	Econometric Purpose
T-0	0 – 24 Hours to Departure	18.2%	2.58 ×	Last-minute emergency / Corporate	Surge surveillance & spot volatility tracking
T-7	2 – 7 Days to Departure	31.4%	1.60 ×	Short-range business & urgent family	High-frequency business cycle sensitivity
T-15	8 – 15 Days to Departure	26.8%	1.25 ×	Planned travel / Seasonal leisure	Balanced consumer welfare benchmark
T-30+	16 – 30+ Days to Departure	23.6%	1.00 × (Baseline)	Advance budget planning / Vacation	Core Structural CPI Inflation Anchor

3.1 Mathematical Formulation of Yield Escalation

$$P(t) = P_{\text{base}} \cdot \left(1 + \alpha \cdot \exp\left(-\lambda \cdot \frac{T_{\text{dep}} - t}{\Delta_{\text{horizon}}}\right)\right) \cdot \left(1 + \eta \cdot \left[\frac{\text{SeatsSold}}{\text{Capacity}}\right]^\gamma\right)$$

Equation 3: Empirical escalation curve with inventory exhaustion dynamics

4. NON-DESTRUCTIVE STATISTICAL ANOMALY & OUTLIER SCREENING ENGINE

Airfares occasionally experience extreme deviations due to regional weather disruptions, festival spikes, algorithm misconfigurations, or web-scraping artifacts. In adherence to strict international statistical guidelines, the platform enforces **non-destructive screening**: observations are never permanently deleted; rather, they are flagged, assigned an audit severity score, and processed via statistical Winsorization.

4.1 Screening Rule 1: Rolling 30-Day Stratified Z-Score

For each route r and lead-time window $w \in \{\text{ext}\{T-0\}, \text{ext}\{T-7\}, \text{ext}\{T-15\}, \text{ext}\{T-30+\}\}$, the historical trailing 30-day mean ($\mu_{r,w}$) and standard deviation ($\sigma_{r,w}$) are updated daily:

$$\begin{aligned} \mu_{r,w,30d} &= \frac{1}{N} \sum_{au=t-30}^{t-1} p_{r,w, au}, \quad \sigma_{r,w,30d} = \\ &\sqrt{\frac{1}{N-1} \sum_{au=t-30}^{t-1} (p_{r,w, au} - \mu_{r,w,30d})^2} \\ Z_{i,r,w,t} &= \frac{p_{i,r,w,t} - \mu_{r,w,30d}}{\sigma_{r,w,30d}} \end{aligned}$$

Equation 4: Stratified historical standard score calculation

Z-Score Boundary	Severity Tier	Automated Action	Operational Treatment
$ Z < 2.00$	NORMAL	Passes directly to index engine	Retained at 100% full weight
$2.00 \leq Z < 3.25$	MEDIUM	Flagged for peer corroboration	Passed with audit tag
$ Z \geq 3.25$	HIGH	Sent to Adjudication Queue	Adjudicated & Winsorized to 95th/5th percentile

4.2 Screening Rule 2: Bilateral Cross-Carrier Peer Divergence

If a single airline's fare spikes while other carriers on the identical corridor remain at baseline, it indicates isolated inventory exhaustion or predatory algorithm surge rather than general market cost inflation:

$$\Delta_{\text{ext}\{\text{peer}\}} = \frac{p_{i,r,t} - \text{Median}_{k \in i}(p_{k,r,t})}{\text{Median}_{k \in i}(p_{k,r,t})}$$

Condition: Flagged if $\Delta_{\text{ext}\{\text{peer}\}} > +1.10$ (+110% divergence above peer median) with remaining seats ≤ 4

4.3 Screening Rule 3: Negative Outlier Floor Test

$$\text{ext}\{\text{Floor Condition:}\} \quad p_{i,r,t} < 0.40 \times \text{Min}_{30d}(p_{r,w})$$

Catches aggregator voucher scraping bugs (e.g. promo discounts erroneously applied before checkout)

4.4 Statistical Treatment: Non-Destructive Winsorization Algorithm

Validated outliers are not purged (which would distort flight volume counts); instead, they are transformed via two-sided Winsorization:

ALGORITHM 1: NON-DESTRUCTIVE ROBUST WINSORIZATION

```
01 Procedure WinsorizeCorridorFares(CorridorFares D, PercentileThreshold  $\alpha = 0.05$ ):
02   P_low ← Percentile(D,  $\alpha$ )
03   P_high ← Percentile(D, 1 -  $\alpha$ )
04   For each observation x in D do:
05     If x.status == "VALIDATED_ANOMALY" then:
06       If x.fare > P_high: x.fare_calc ← P_high; x.treatment ← "TOP_WINSORIZED"
07       Else if x.fare < P_low: x.fare_calc ← P_low; x.treatment ← "BOTTOM_WINSORIZED"
08       End if
09     Else: x.fare_calc ← x.fare; End if
10   End for
11   Return D
```

5. MATHEMATICAL AIRFARE PRICE INDEX (API) FORMULATION

The core computational engine aggregates cleaned and Winsorized observations into high-level headline indices. In compliance with MoSPI standards and the International Labour Organization (ILO) Consumer Price Index Manual, index compilation occurs in a two-stage hierarchical process.

5.1 Stage I: Elementary Aggregates (Observation Level)

At the unweighted flight observation level within corridor r and advance window w , two elementary index formulas are maintained:

$$\text{Jevons Elementary Index: } I_J^{(t,0)} = \left(\prod_{i=1}^n \frac{p_{i,t}}{p_{i,0}} \right)^{\frac{1}{n}} = \exp \left(\frac{1}{n} \sum_{i=1}^n \ln \frac{p_{i,t}}{p_{i,0}} \right)$$

Geometric mean of price relatives. Satisfies Axiom of Transitivity and avoids upward arithmetic bias.

$$\text{Dutot Elementary Index: } I_D^{(t,0)} = \frac{\frac{1}{n} \sum_{i=1}^n p_{i,t}}{\frac{1}{n} \sum_{i=1}^n p_{i,0}}$$

Ratio of arithmetic average prices. Used as cross-check against Jevons to identify price variance dispersion.

5.2 Stage II: Passenger-Weighted Route Basket Aggregation

Corridors are weighted according to official DGCA passenger volume shares W_r , where $\sum_{r=1}^M W_r = 1.0$. The platform monitors three distinct tier sub-indices:

- **Metro-Metro Corridors (48.0% National Weight):** High density (DEL-BOM, BLR-DEL, BOM-MAA) with corporate traffic.
- **Metro-to-Non-Metro Corridors (34.0% National Weight):** Seasonal tourist and trade routes (BOM-GOI, DEL-PAT, CCU-GAU).
- **UDAN Regional Connectivity Corridors (18.0% National Weight):** Government-subsidized regional connectivity scheme routes.

5.3 Higher-Level Composite Formulae

$$\text{Laspeyres Price Index: } I_L^{(t,0)} = \frac{\sum_{r=1}^M p_{r,t} \cdot q_{r,0}}{\sum_{r=1}^M p_{r,0} \cdot q_{r,0}} = \sum_{r=1}^M W_{r,0} \cdot \left(\frac{p_{r,t}}{p_{r,0}} \right)$$

Fixed base-period volume weighted. Standard international benchmark, but subject to upward substitution bias.

$$\text{Paasche Price Index: } I_P^{(t,0)} = \frac{\sum_{r=1}^M p_{r,t} \cdot q_{r,t}}{\sum_{r=1}^M p_{r,0} \cdot q_{r,t}}$$

Current-period volume weighted. Reflects real-time traveler substitution, but requires live ticketing load factors.

$$\text{Fisher Ideal Superlative Index: } I_F^{(t,0)} = \sqrt{I_L^{(t,0)} \cdot I_P^{(t,0)}}$$

Official MoSPI Recommended Methodology. Geometric mean of Laspeyres and Paasche. Passes both Time Reversal and Factor Reversal Axioms.

6. HIERARCHICAL MOVEMENT DECOMPOSITION & ATTRIBUTION ENGINE

A key regulatory requirement for MoSPI and DGCA is **explainability**: whenever the headline national index moves (e.g. \$+3.24\$ points MoM), policymakers must understand precisely which corridor, which carrier, and which booking lead-time window caused the movement.

6.1 Additive Log-Mean Index Decomposition

The platform implements exact additive decomposition using the Montgomery-Bortkiewicz theorem, guaranteeing that the sum of all lower-level point contributions equals 100.0% of the total index movement:

$$\begin{aligned} \Delta I_F &= I_F^{(t)} - I_F^{(t-1)} = \sum_{r=1}^M C_r \\ C_r &= \frac{1}{2} \cdot \left[W_r \cdot \left(\frac{p_{r,t}}{p_{r,0}} - \frac{p_{r,t-1}}{p_{r,0}} \right) \cdot \sqrt{\frac{I_P^{(t)}}{I_L^{(t)}}} + \Delta C_{\text{Paasche},r} \right] \end{aligned}$$

Equation 5: Additive corridor contribution satisfying exact mathematical closure ($\sum C_r \equiv \Delta I_F$)

Each corridor contribution C_r is recursively decomposed down through the attribution hierarchy:

$$C_r = \sum_{a \in \text{Airlines}} C_{r,a} = \sum_{a \in \text{Airlines}} \sum_{w \in \{ \text{T-0, T-7, T-15, T-30} \}} C_{r,a,w}$$

Equation 6: Multi-tiered recursive attribution (Corridor → Carrier → Horizon)

7. CRYPTOGRAPHIC LINEAGE & IMMUTABLE AUDIT TRAIL (SHA-256)

To prevent data tampering and guarantee total statistical reproducibility for judicial or parliamentary inquiries, each execution run generates an immutable audit record linked via a SHA-256 cryptographic hash chain:

$$\mathcal{H}_k = \text{SHA-256}(\mathcal{H}_{k-1}, \text{RunID}_k, \text{StageID}_k, \text{FormulaVer}_k, N_{\text{in}}, N_{\text{out}}, \text{Timestamp}_k)$$

Equation 7: Consecutive cryptographic block linkage across data stages

AUDIT STAGE EXECUTION SEQUENCE

- S1 Stage 1: Multi-Source Scraper Coordinator → 149,020 raw observations harvested.
- S2 Stage 2: Cleaning & Normalization → 730 duplicates purged, statutory taxes segregated.
- S3 Stage 3: Lead-Time Stratification → Partitioned into T-0 (18%), T-7 (31%), T-15 (27%), T-30+ (24%).
- S4 Stage 4: Anomaly Scoring Engine → 14 high outliers flagged; non-destructive Winsorization applied.
- S5 Stage 5: Statistical Index Engine → Fisher Composite calculated: 118.42 (+3.24% MoM). Hash sealed.

Audit Guarantee: Any modification to raw scraper outputs or calculation weights breaks hash consistency $\mathcal{H}_k$, immediately triggering an automated alert in the MoSPI Audit Center.

8. PARAMETER BENCHMARKS & ALGORITHMIC COMPLEXITY

Module	Algorithm / Equation	Default Threshold	Time Complexity	Execution Frequency
Tax Normalization	Equation 1: Fee Stripping	Statutory rate table	$\mathcal{O}(N)$	Stream (Real-time)
Collision Resolution	Equation 2: SHA-256 Hash Key	30-minute deduplication slot	$\mathcal{O}(N \log N)$	Hourly batch
Lead-Time Binning	Days to departure ($T_{\text{ext}\{\text{dep}\}} - t$)	4 bins: 0-1d, 2-7d, 8-15d, 16-30d+	$\mathcal{O}(N)$	Real-time pipeline
Anomaly Detection	Equation 4: Stratified Z-Score	$ Z \geq 2.50$ (Med), $ Z \geq 3.25$ (High)	$\mathcal{O}(N)$	Hourly evaluation
Winsorization	Algorithm 1: Two-Sided Trim	$\alpha = 0.05$ (5th / 95th Percentile)	$\mathcal{O}(N \log N)$	Pre-aggregation step
Elementary Index	Jevons Geometric Mean	Lower observation aggregate	$\mathcal{O}(M \cdot n)$	Daily & Monthly
Composite Index	Fisher Ideal ($I_F = \sqrt{I_L \cdot I_P}$)	Base year 2024 = 100.0	$\mathcal{O}(M)$	Daily & Monthly
Attribution Tree	Equation 5-6: Log-Mean Decomposition	Exact closure ($\sum = 100\%$)	$\mathcal{O}(M \cdot K)$	On-demand / Monthly
Audit Trail	Equation 7: SHA-256 Merkle Link	Cryptographic hash block	$\mathcal{O}(1)$ per stage	Pipeline stage completion